

Lab 03: DNS & DHCP Servers

Automating IP Assignment and Name Resolution

Lab #3

Topic: Network Services

Tool: Cisco Packet Tracer

Cert: CompTIA Network+

1. Lab Objectives

- Configure a DHCP server to automatically assign IP addresses to client PCs.
- Configure a DNS server to resolve hostnames to IP addresses.
- Verify DHCP lease acquisition and DNS name resolution from client machines.
- Understand how DHCP reduces administrative overhead in large networks.
- Relate to Network+ objectives: 1.6 – Explain the use of DNS and DHCP.

2. Network Topology

Two servers (DHCP and DNS) are connected via a switch to client PCs and a router. The DHCP server hands out addresses in the 192.168.20.0/24 range. The DNS server is configured at 192.168.20.5 with static hostname records.

Device	Role	IP Address	Notes
Router0	Default Gateway	192.168.20.1	Static
DHCP-Server	DHCP Server	192.168.20.2	Static
DNS-Server	DNS Server	192.168.20.5	Static
PC0	DHCP Client	Dynamic	From pool
PC1	DHCP Client	Dynamic	From pool

3. Step-by-Step Configuration

3.1 DHCP Server Configuration

Click DHCP-Server → Services → DHCP. Enable the service and create a pool:

Pool Name	LAN_POOL
Default GW	192.168.20.1
DNS Server	192.168.20.5

Start IP	192.168.20.50
Subnet Mask	255.255.255.0
Max Users	50

3.2 DNS Server Configuration

Click DNS-Server → Services → DNS. Enable DNS and add A records:

Hostname (Name)	Record Type	IP Address
dhcp-server.lab	A Record	192.168.20.2
dns-server.lab	A Record	192.168.20.5
router.lab	A Record	192.168.20.1

3.3 Configure PCs as DHCP Clients

PC0/PC1 → Desktop → IP Configuration:

Select: DHCP (instead of Static)

Wait for lease: IP assigned from pool (e.g. 192.168.20.50)

GW and DNS should auto-populate from server options

4. Verification

PC0> ipconfig ! Confirm DHCP lease received

PC0> ping 192.168.20.5 ! Ping DNS server by IP

PC0> ping dns-server.lab ! Test DNS name resolution

PC0> ping router.lab ! Confirm DNS resolves gateway name

5. Key Takeaways

- DHCP uses a 4-step DORA process: Discover → Offer → Request → Acknowledge.
- Static IPs should be assigned to servers, routers, and printers — DHCP is best for clients.
- DNS translates human-readable hostnames (e.g. router.lab) to IP addresses.
- The DNS server IP is distributed to clients via DHCP option 6 automatically.
- Without DNS, users must know exact IP addresses — a major usability limitation.